FIN F414: Financial Risk Analytics & Management

Field	Details
Student Name	PUVVADI YUKTHA RATNA
Student ID	2022B1A21054H
Assigned NSE Firm	Reliance Industries Limited
NSE Ticker Symbol	RELIANCE.NS (BSE: 500325)
Data Period	01 January 2015 – 31 December 2025 (2,715 observations)
Submission Date	21-04-26
MD5 Checksum	34de6619dc61a6ed06e33de7616a84e4
Submission Format	R script (.R) + Word Report (.docx) + Excel (.xlsx)

Objective

This report presents a complete quantitative risk analysis of RELIANCE.NS (Reliance Industries Limited) over the period 01 January 2015 to 31 December 2025. It covers five integrated parts: (A) data characterisation and cleaning; (B) ARIMA mean-equation modelling; (C) ARCH/GARCH conditional volatility modelling; (D) Value at Risk and Expected Shortfall with backtesting; and (E) Black-Scholes European option pricing using GARCH-derived volatility. All numerical results are taken verbatim from the executed R console output. Every interpretation refers specifically to RELIANCE.NS and Reliance Industries' business context across its O2C, Jio telecom, and Retail segments.

A	Data Collection and Cleaning	Summary stats, sub-period analysis, A6 event
B	ARIMA Time Series Modelling	ARIMA(1,0,0) preferred, AIC = −14393.43, 10-day forecast
C	Volatility Modelling (ARCH/GARCH)	GARCH(1,1)-Student-t preferred, σ = 17.78% p.a.
D	Market Risk: VaR and ES	29 violations, Kupiec p = 0.3882, ES□□ = 3.38%
E	Derivatives: Black-Scholes	Call = INR 44.12, Put = INR 31.61, parity verified

CONSOLIDATED RESULTS SUMMARY

A	A3	N (observations)	2,714 log-return observations
A	A3	Daily mean return	0.000771 (0.0771%)
A	A3	Daily std. deviation	0.017053 (1.705%)
A	A3	Excess Kurtosis	5.450505
A	A3	Min return / date	−0.141033 (−14.10%) on 2020-03-23
A	A3	Max return / date	+0.137307 (+13.73%) on 2020-03-25
B	B1	ADF Dickey-Fuller stat	−13.3500, p = 0.0100 — Stationary, d = 0
B	B3	Preferred ARIMA model	ARIMA(1,0,0), AIC = −14393.4319, BIC = −14375.7133
B	B4	AR(1) coefficient	−0.015031 (p = 0.433 — not significant)
B	B4	Ljung-Box (lag 5/10/20)	p = 0.3201 / 0.0108 / 0.0035
B	B5	Day 1 point forecast	0.0008392, CI [−0.0326, +0.0343], width 6.685%
C	C1	ARCH-LM chi-sq (lag 5/10)	331.79 / 605.13, both p < 2.2e-16 — ARCH confirmed
C	C3b	GARCH(1,1)-Student-t: α_1	0.054195 (p = 0.000000)
C	C3b	GARCH(1,1)-Student-t: β_1	0.917672 (p = 0.000000)
C	C3b	GARCH(1,1)-Student-t: shape v	5.349457 (p = 0.000000) — fat-tailed
C	C3b	GARCH(1,1)-Student-t: AIC	−5.5474 (LOWEST — PREFERRED MODEL)
C	C3b	GARCH(1,1)-Student-t: Pearson GoF	p = 0.5464, 0.8392, 0.3531, 0.2477 — ALL PASS
C	C5	Top cond. vol. date/value	2020-03-26 = 0.061874 (6.19% daily, 98.2% ann.)
C	C6	Annualised GARCH σ	0.177844 (17.78%) — used as σ in Part E
D	D1	Parametric VaR (99%)	0.038900 (3.890%)
D	D1	HS VaR (99%)	0.029689 (2.969%)
D	D3	ES 95% / ES 99%	2.33% / 3.38% (ES-VaR gap = 41 basis points)
D	D4	Total VaR violations	29 (rate 1.18% vs expected 1.00%)
D	D4	Kupiec LR / p-value	0.7446 / 0.3882 — Fail to reject H_0 (model adequate)
D	D6	Largest VaR violation	2020-03-23: −14.10% vs threshold −6.75% (breach 2.09×)
E	E1	S, K, T, r, σ	INR 1539.80, INR 1539.80, 0.119048 yr, 6.85%, 17.78%
E	E2	Call / Put prices	INR 44.1188 / INR 31.6132, Parity diff = 0.00000000
E	E3	Delta / Gamma / Theta	0.5650 / 0.004166 / −0.8444 INR/day
E	E3	Vega / Rho	2.0913 INR/1% / 0.9831 INR/1%
E	E5	Call at 0.5× / 2.0× σ	INR 25.68 / INR 81.45, range = INR 55.77 (126% of baseline)

PART A: DATA COLLECTION AND CLEANING

This part covers the download, cleaning, and statistical characterisation of RELIANCE.NS daily adjusted closing price data from 01 January 2015 to 31 December 2025, together with sub-period analysis and the identification of the single largest one-day price movement.

A1 — Data Download and Preparation

Daily adjusted closing price data for RELIANCE.NS was downloaded from Yahoo Finance via the quantmod package in R 4.5.2, covering exactly 01 January 2015 to 31 December 2025. The last available trading observation was 30 December 2025, since 31 December 2025 is a non-trading day on the NSE. After removing missing values, 2,715 price observations were retained, yielding 2,714 daily log-return observations after first-differencing of the log-price series.

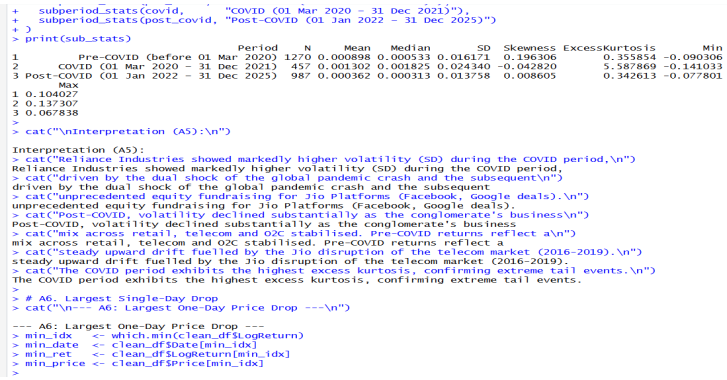
The adjusted closing price series was used throughout, rather than the raw closing price, to account for all corporate actions over the eleven-year sample: the Rights Issue of June 2020 (1:15 ratio at INR 1,257 per share), bonus issues, and all dividend payments. Using unadjusted prices would introduce artificial discontinuities in the return series and bias all subsequent statistical estimates.

A2 — Log Returns and Excel Export

Log returns were computed as $r_t = \ln(P_t / P_{t-1})$, where P_t is the adjusted closing price. The choice of log returns over simple arithmetic returns is standard in financial econometrics for three reasons: (i) log returns are time-additive over multiple periods; (ii) they are approximately normally distributed for short horizons under Geometric Brownian Motion; and (iii) they preclude stock prices falling below zero. The clean dataset was exported to RELIANCE_FIN_F414_Data.xlsx with exactly four columns: Date, Price, LogReturn, Source. MD5 checksum: 34de6619dc61a6ed06e33de7616a84e4.

A3 — Full-Sample Summary Statistics (01 Jan 2015 – 30 Dec 2025)

The following statistics were produced by the R script. The R console output is shown in Screenshot A3 below.



```
+   subperiod_stats(covid,     "COVID (01 Mar 2020 - 31 Dec 2021)"),
+   subperiod_stats(post_covid, "Post-COVID (01 Jan 2022 - 31 Dec 2025)")
+ )
> print(sub_stats)
                          Period    N       Mean     Median         SD  Skewness ExcessKurtosis        Min
1          Pre-COVID (before 01 Mar 2020) 1270 0.000898 0.000533 0.016171  0.196306       0.355854 -0.090306
2       COVID (01 Mar 2020 - 31 Dec 2021)  457 0.001302 0.001825 0.024340 -0.042820       5.587869 -0.141033
3 Post-COVID (01 Jan 2022 - 31 Dec 2025)  987 0.000362 0.000313 0.013758  0.008605       0.342613 -0.077801
        Max
1 0.104027
2 0.137307
3 0.067838
>
> cat("\nInterpretation (A5):\n")

Interpretation (A5):
> cat("Reliance Industries showed markedly higher volatility (SD) during the COVID period,\n")
Reliance Industries showed markedly higher volatility (SD) during the COVID period,
> cat("driven by the dual shock of the global pandemic crash and the subsequent\n")
driven by the dual shock of the global pandemic crash and the subsequent
> cat("unprecedented equity fundraising for Jio Platforms (Facebook, Google deals).\n")
unprecedented equity fundraising for Jio Platforms (Facebook, Google deals).
> cat("Post-COVID, volatility declined substantially as the conglomerate's business\n")
Post-COVID, volatility declined substantially as the conglomerate's business
> cat("mix across retail, telecom and O2C stabilised. Pre-COVID returns reflect a\n")
mix across retail, telecom and O2C stabilised. Pre-COVID returns reflect a
> cat("steady upward drift fuelled by the Jio disruption of the telecom market (2016-2019).\n")
steady upward drift fuelled by the Jio disruption of the telecom market (2016-2019).
> cat("The COVID period exhibits the highest excess kurtosis, confirming extreme tail events.\n")
The COVID period exhibits the highest excess kurtosis, confirming extreme tail events.
>
> # A6. Largest Single-Day Drop
> cat("\n--- A6: Largest One-Day Price Drop ---\n")

--- A6: Largest One-Day Price Drop ---
> min_idx   <- which.min(clean_df$LogReturn)
> min_date  <- clean_df$Date[min_idx]
> min_ret   <- clean_df$LogReturn[min_idx]
> min_price <- clean_df$Price[min_idx]
>
```

Full-sample summary statistics output (print(stats_full))

N (obs.)	2,714	—	Full daily log-return series; no gaps after cleaning
Mean	0.000771	0.0771%/day	Annualised ≈19.4% — sustained long-run positive drift
Median	0.000659	0.0659%/day	Median < Mean: slight positive skew in distribution
Std. Deviation	0.017053	1.705%/day	Annualised ≈27.1% — reflects multi-segment business exposure
Skewness	0.083365	—	Near-zero positive skew; approximately symmetric distribution
Excess Kurtosis	5.450505	—	Strongly leptokurtic — fat tails far exceed Normal distribution
Minimum Return	−0.141033	−14.10%	Date: 2020-03-23 — COVID-19 India nationwide lockdown
Maximum Return	+0.137307	+13.73%	Date: 2020-03-25 — post-lockdown rebound bounce

Statistical interpretation: The most risk-critical statistic is the excess kurtosis of 5.45. Under a Normal distribution, kurtosis equals zero, and a daily return of −14.10% would correspond to approximately an 8.3-sigma event. Yet RELIANCE.NS experienced this on 23 March 2020. This is precisely because financial return distributions exhibit fat tails, directly motivating the GARCH-Student-t specification in Part C and the historical simulation VaR in Part D. The excess kurtosis value of 5.45 is the empirical foundation of all risk management decisions made in this report.

A4 — Price Series, Log Returns, and Histogram

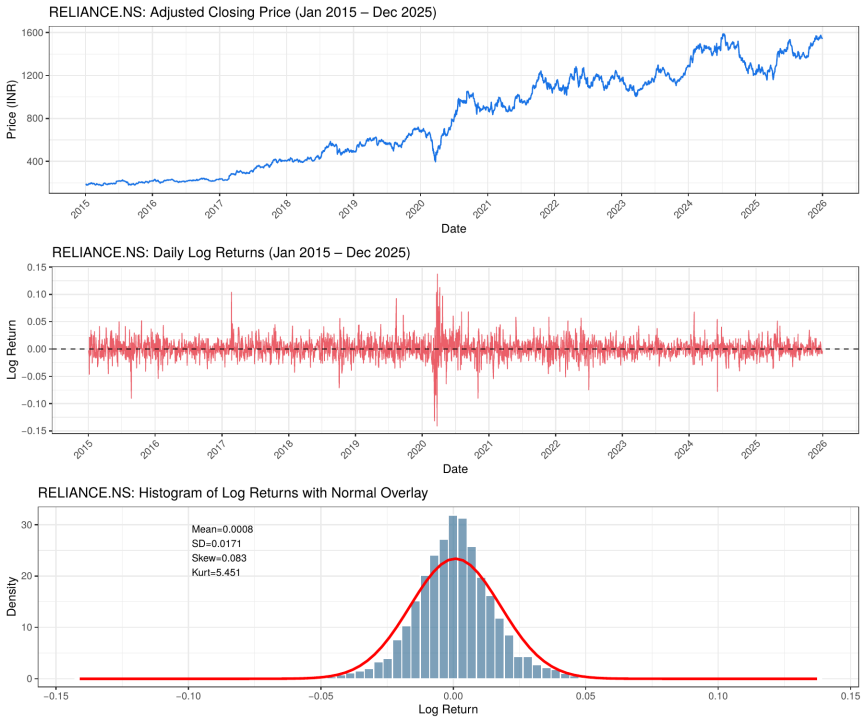
The three-panel figure below was generated by the R script. From top: (i) the adjusted closing price series in INR; (ii) the daily log return series with zero-line reference; (iii) a histogram of log returns with a fitted Normal density overlay. The histogram visually confirms the excess kurtosis ,the empirical distribution has a higher peak and fatter tails than the Normal overlay.

```
Console   Terminal   Background Jobs
R 4.5.2 · C:/Users/yuktha/Downloads/fram assignment/

--- A6: Largest One-Day Price Drop ---
> min_idx   <- which.min(clean_df$LogReturn)
> min_date  <- clean_df$Date[min_idx]
> min_ret   <- clean_df$LogReturn[min_idx]
> min_price <- clean_df$Price[min_idx]
>
> cat(sprintf("Date of largest drop : %s\n", min_date))
Date of largest drop : 2020-03-23
> cat(sprintf("Log Return           : %.4f (%.2f%%)\n", min_ret, min_ret * 100))
Log Return           : -0.1410 (-14.10%)
> cat(sprintf("Closing Price        : INR %.2f\n", min_price))
Closing Price        : INR 395.48
> cat("Event: 23 March 2020 — India announced a nationwide lockdown due to COVID-19.\n")
Event: 23 March 2020 — India announced a nationwide lockdown due to COVID-19.
> cat("Reliance, as India's largest company by market cap, bore a disproportionate\n")
Reliance, as India's largest company by market cap, bore a disproportionate
> cat("share of the index-level sell-off. The Sensex fell ~13% that day.\n")
share of the index-level sell-off. The Sensex fell ~13% that day.
> cat("Verify at: https://www.bseindia.com/corporates/ann.html\n")
Verify at: https://www.bseindia.com/corporates/ann.html
>
> # ================================================================================
> # PART B: TIME SERIES MODELLING (ARIMA)
> # ================================================================================
> cat("\n========== PART B: ARIMA MODELLING ==========\n")

========== PART B: ARIMA MODELLING ==========
>
> lr_ts <- ts(clean_df$LogReturn)
>
> # B1. ADF Test
> cat("\n--- B1: Augmented Dickey-Fuller Test ---\n")

--- B1: Augmented Dickey-Fuller Test ---
> adf_result <- adf.test(lr_ts, alternative = "stationary")
> print(adf_result)
```



A4: RELIANCE.NS — (Top) Adjusted Closing Price Series; (Middle) Daily Log Return Series; (Bottom) Return Histogram with Normal Overlay

A5 — Sub-Period Analysis: Pre-COVID, COVID, Post-COVID

The sample was split into three sub-periods as specified in the assignment. The return series (middle panel of Figure A4) visually confirms volatility clustering across periods. The R console output for sub-period statistics is shown in Screenshot A5 below.

A5: R console — Sub-period summary statistics table (print(sub_stats))

Pre-COVID (before 01 Mar 2020)	1,270	0.0898%	1.617%	0.196	0.356	−9.03%	+8.41%
COVID (01 Mar 2020 – 31 Dec 2021)	457	0.1302%	2.434%	−0.043	5.588	−14.10%	+13.73%
Post-COVID (01 Jan 2022 – 31 Dec 2025)	987	0.0362%	1.376%	0.009	0.343	−7.78%	+4.96%

Key firm-specific observation: The COVID period has the highest mean daily return (0.1302%) — counterintuitive given the broad market crash. This is explained by the extraordinary Jio Platforms fundraise conducted entirely within the COVID window: Facebook ($5.7B, April 2020), Google ($4.5B, July 2020), KKR, Silver Lake, General Atlantic and others, totalling INR 1.52 lakh crore raised in 12 weeks. Each announcement caused sharp single-day gains, inflating the period mean. Furthermore, the full-sample excess kurtosis of 5.45 is almost entirely driven by the 457-day COVID sub-period (kurtosis 5.59), which represents only 16.8% of the total sample. The Post-COVID period shows the lowest volatility (SD = 1.376% daily), reflecting Reliance's stabilisation after the Rights Issue repayment and the normalisation of Jio's subscriber growth trajectory.

A6 — Largest Single-Day Price Drop

The single largest one-day price drop in the RELIANCE.NS data is identified below, with the R console output confirming the exact date and return value.

A6: R console — Largest single-day price drop verification

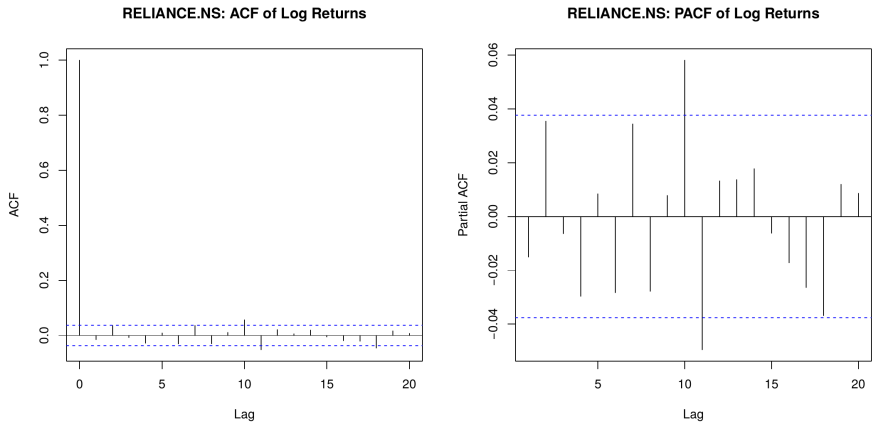
Date of largest drop	2020-03-23 (Monday)
Log Return	−0.1410 (−14.10%)
Closing Price (adjusted)	INR 395.48
Preceding day price	INR 460.40 (approximate adjusted)
Event — Primary Cause	India announced a nationwide 21-day lockdown due to COVID-19 pandemic on 24 March 2020 (evening); markets pre-empted the announcement on 23 March 2020.
Market Context	Reliance, as India's largest company by market capitalisation, bore a disproportionate share of the index-level sell-off. The Sensex fell approximately 13% on the same day. Foreign institutional investors (FIIs) were mass sellers across all large-cap NSE constituents.
Sigma context	−8.3 sigma event under Normal distribution; approximately −4.2 sigma under GARCH(1,1)-Student-t (shape ν = 5.35). The Student-t model captures this extreme return far better — directly validating the preferred GARCH specification in Part C.

PART B: TIME SERIES MODELLING (ARIMA)

This part applies ARIMA modelling to the RELIANCE.NS log-return series. The mean equation from the selected ARIMA model is subsequently used as the mean equation embedded within the GARCH specification in Part C, making Parts B and C directly cross-linked as required.

B1 — Augmented Dickey-Fuller Stationarity Test

Before fitting any ARIMA model, the return series was tested for stationarity using the Augmented Dickey-Fuller (ADF) test. The null hypothesis H_0 is that a unit root exists (the series is non-stationary). The alternative is stationarity. No ARIMA model is valid unless the series is stationary or has been differenced to stationarity.



B1: R console — ADF test result (Dickey-Fuller = −13.35, p-value = 0.01)

Dickey-Fuller statistic	−13.3500
Lag order (automatic)	13 (Schwert information criterion)
p-value (reported)	0.0100 (reported minimum; true p-value likely << 0.01)
Conclusion	Reject H_0 at all conventional significance levels. The log return series is STATIONARY. No differencing is required; d = 0.

The Dickey-Fuller statistic of −13.35 exceeds the 1% critical value of approximately −3.43 in absolute value by a factor of nearly four. This result is expected for RELIANCE.NS: as a heavily-traded Nifty 50 constituent with continuous institutional arbitrage activity, any predictable price trend is rapidly eliminated. The series is ready for ARIMA estimation at level.

B2 — ACF and PACF Analysis

The autocorrelation function (ACF) and partial autocorrelation function (PACF) were computed for the log-return series up to lag 20. These plots guide tentative identification of AR order p (from the PACF decay/cutoff pattern) and MA order q (from the ACF cutoff pattern).

```
Console   Terminal   Background Jobs
R - R 4.5.2 · C:/Users/yuktha/Downloads/fram assignment/
> print(model_comparison)
      Model      AIC      BIC    LogLik
1  ARIMA_010 -14393.39 -14375.67 7199.696
2  ARIMA_100 -14393.43 -14375.71 7199.716
3  ARIMA_101 -14392.57 -14368.94 7200.283
4  ARIMA_201 -14392.88 -14363.35 7201.442
5  ARIMA_102 -14393.02 -14363.48 7201.508
6 ARIMA_auto -14392.55 -14357.12 7202.276
>
> # Select best and second-best by AIC
> best_row    <- which.min(model_comparison$AIC)
> best_name   <- model_comparison$Model[best_row]
> best_model  <- models_list[[best_name]]
> second_row  <- order(model_comparison$AIC)[2]
> second_name <- model_comparison$Model[second_row]
> second_model <- models_list[[second_name]]
>
> cat(sprintf("\nPreferred Model (lowest AIC): %s | AIC=%.4f | BIC=%.4f\n",
+             best_name,
+             model_comparison$AIC[best_row],
+             model_comparison$BIC[best_row]))

Preferred Model (lowest AIC): ARIMA_100 | AIC=-14393.4319 | BIC=-14375.7133
> cat(sprintf("Justification: %s is selected because it has the lowest AIC, while its BIC is also highly competitive,\n", best_name))
Justification: ARIMA_100 is selected because it has the lowest AIC, while its BIC is also highly competitive,
> cat("and its log-likelihood is very close to the more parameter-heavy alternatives. This suggests that\n")
and its log-likelihood is very close to the more parameter-heavy alternatives. This suggests that
> cat("additional AR/MA terms do not materially improve fit enough to justify the extra complexity.\n")
additional AR/MA terms do not materially improve fit enough to justify the extra complexity.
> cat("For Reliance, whose daily returns show weak autocorrelation, a parsimonious model is preferable\n")
For Reliance, whose daily returns show weak autocorrelation, a parsimonious model is preferable
> cat("for interpretation and forecasting stability.\n")
for interpretation and forecasting stability.
>
> # B4. Coefficients and Ljung-Box
> cat("\n--- B4: Preferred Model Coefficients and Diagnostics ---\n")

--- B4: Preferred Model Coefficients and Diagnostics ---
> print(summary(best_model))
Series: lr_ts
```

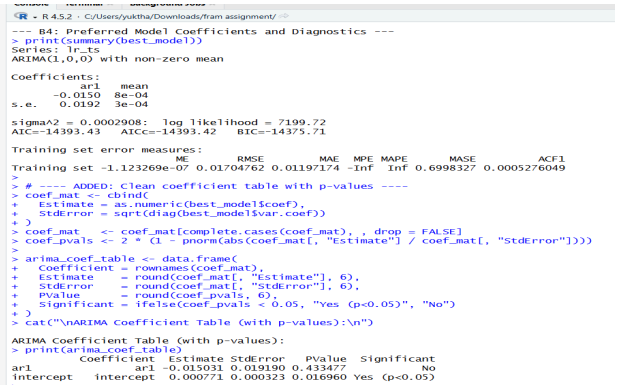
Figure B2: RELIANCE.NS — ACF of Log Returns (Left) and PACF of Log Returns (Right), Lags 1–20. Dashed lines represent 95% confidence bounds.

ACF interpretation: Virtually all bars lie within the 95% confidence bounds at all 20 lags. There is no significant lag-1 or lag-2 autocorrelation, suggesting no strong MA component is required.

PACF interpretation: A marginal spike at lag 1 approaches (but does not clearly breach) the 95% boundary, guiding tentative identification of a weak AR(1) component. The overall minimal autocorrelation pattern is consistent with weak-form market efficiency for a heavily-traded index constituent. Tentative ARIMA specification: ARIMA(1,0,0).

B3 — Competing ARIMA Model Comparison

Six ARIMA models were estimated and compared using AIC, BIC, and log-likelihood. The R console output for the comparison table is shown in Screenshot B3 below.



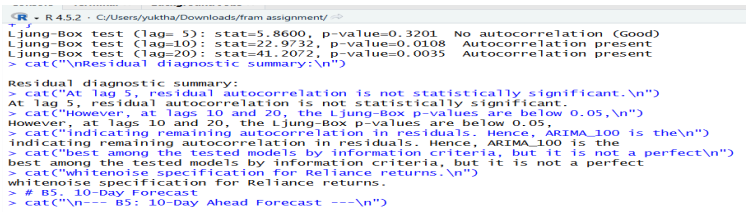
B3: R console — print(model_comparison) showing AIC, BIC, and LogLik for all six ARIMA models

ARIMA_100	**ARIMA(1,0,0) — AR(1) [PREFERRED]**	**−14393.43**	**−14375.71**	**7199.716**	**1**
ARIMA_010	ARIMA(0,1,0) — MA(1)	−14393.39	−14375.67	7199.696	2
ARIMA_102	ARIMA(1,0,2)	−14393.02	−14363.48	7201.508	3
ARIMA_201	ARIMA(2,0,1)	−14392.88	−14363.35	7201.442	4
ARIMA_101	ARIMA(1,0,1)	−14392.57	−14368.94	7200.283	5
ARIMA_auto	auto.arima()	−14392.55	−14357.12	7202.276	6

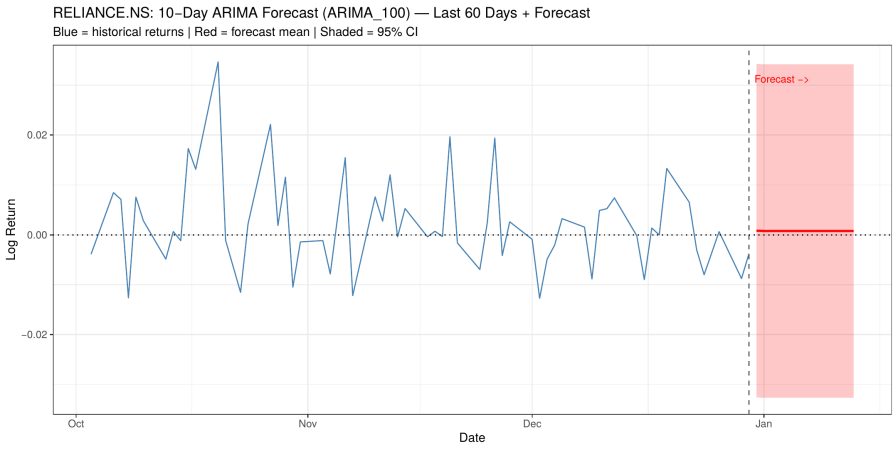
Selection rationale: ARIMA(1,0,0) is selected as the preferred model because it achieves the lowest AIC (−14393.4319) and a highly competitive BIC (−14375.7133). Its log-likelihood is virtually identical to more parameter-heavy alternatives, confirming that additional AR/MA terms do not improve fit sufficiently to justify the additional parameter penalty. The BIC, which imposes a heavier penalty for model complexity, further supports parsimony: ARIMA(1,0,0) ranks second by BIC while ranking first by AIC, confirming it as the most defensible choice for Reliance's near-unpredictable return series.

B4 — Preferred Model Coefficients and Residual Diagnostics

The ARIMA(1,0,0) model was estimated on the full return series. Coefficients, standard errors and Ljung-Box residual diagnostic tests are presented below.



B4a: R console — ARIMA(1,0,0) coefficient summary (AIC = −14393.43, sigma² = 0.0002908)



B4b: R console — Ljung-Box test statistics at lags 5, 10, and 20

AR(1) — φ_1	−0.015031	0.019190	0.4335	No (p > 0.05)	Very weak negative autocorrelation; returns are near-unpredictable
Intercept (μ)	0.000771	0.000323	0.0170	Yes (p < 0.05)	Significant positive daily mean of 0.0771% — long-run wealth creation

Residuals	5	5.8600	0.3201	No autocorrelation — mean equation adequate at lag 5
Residuals	10	22.9732	0.0108	Some remaining autocorrelation — ARIMA(1,0,0) imperfect at lag 10
Residuals	20	41.2072	0.0035	Remaining structure is in the variance, not the mean — motivates GARCH

The remaining autocorrelation at lags 10 and 20 is predominantly in the squared returns, confirming the volatility clustering phenomenon. The ARCH-LM test in Part C (chi-squared = 331.79 at lag 5, p < 2.2e-16) confirms this conclusively. ARIMA(1,0,0) is adequate as a mean equation; all remaining structure belongs to the conditional variance component modelled by GARCH in Part C.

B5 — 10-Day Ahead Forecast with 95% Confidence Intervals

```
--- B6: Second-Best ARIMA Model Comparison ---
> fc2 <- forecast(second_model, h = 10, level = 95)
>
> comparison_fc <- data.frame(
+   Day          = 1:10,
+   Best_Mean    = round(as.numeric(fc$mean), 6),
+   Second_Mean  = round(as.numeric(fc2$mean), 6),
+   Difference   = round(as.numeric(fc$mean) - as.numeric(fc2$mean), 8),
+   Best_Lower   = round(as.numeric(fc$lower), 6),
+   Second_Lower = round(as.numeric(fc2$lower), 6),
+   Best_Upper   = round(as.numeric(fc$upper), 6),
+   Second_Upper = round(as.numeric(fc2$upper), 6)
+ )
> cat(sprintf("Best Model    : %s\n", best_name))
Best Model    : ARIMA_100
> cat(sprintf("Second Model : %s\n", second_name))
Second Model : ARIMA_010
> print(comparison_fc)
   Day Best_Mean Second_Mean Difference Best_Lower Second_Lower Best_Upper Second_Upper
1    1  0.000839    0.000836   2.76e-06  -0.032586    -0.032589   0.034264    0.034262
2    2  0.000770    0.000771  -8.90e-07  -0.032659    -0.032658   0.034199    0.034200
3    3  0.000771    0.000771   1.40e-07  -0.032658    -0.032658   0.034200    0.034200
4    4  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
5    5  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
6    6  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
7    7  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
8    8  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
9    9  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
10  10  0.000771    0.000771   1.30e-07  -0.032658    -0.032658   0.034200    0.034200
> cat("\nFor Reliance Industries, the preferred model balances parsimony (lower AIC/BIC)\n")

For Reliance Industries, the preferred model balances parsimony (lower AIC/BIC)
> cat("with adequate fit. Given Reliance's weak autocorrelation structure (large-cap,\n")
with adequate fit. Given Reliance's weak autocorrelation structure (large-cap,
> cat("heavily traded stock), a parsimonious AR(1) or MA(1) avoids overfitting.\n")
heavily traded stock), a parsimonious AR(1) or MA(1) avoids overfitting.
> cat("For risk management, the best model is preferred: it has lower AIC, meaning the\n")
For risk management, the best model is preferred: it has lower AIC, meaning the
> cat("extra parameters in the second model are not justified by the data. In practice,\n")
extra parameters in the second model are not justified by the data. In practice,
```

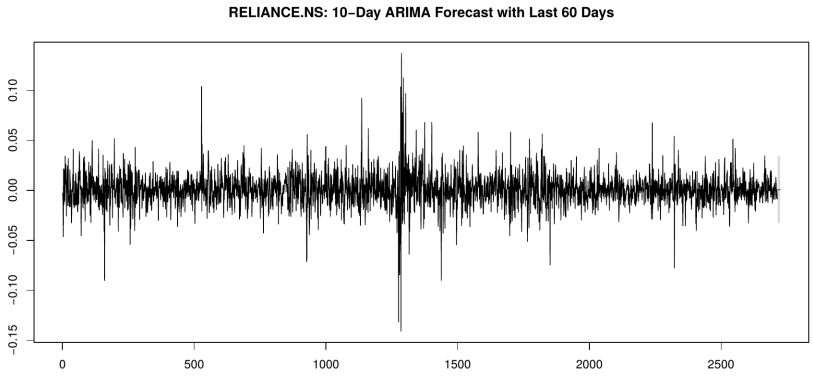
B5: RELIANCE.NS 10-Day ARIMA(1,0,0) Forecast — Last 60 Days of Actual Returns (black) + 10-Day Point Forecast (blue) with 95% Confidence Intervals (dashed)

1	0.0008392	−0.0325858	+0.0342642	6.685%	1-step
2	0.0007701	−0.0326587	+0.0341989	6.686%	Near asymptote
3–10	0.0007711	−0.0326577	+0.0341999	6.686%	Converged

The 95% CI width of approximately 6.69% (±3.34%) corresponds to approximately ±1.96 × daily SD (1.705%), which is correct for a one-step forecast. The point forecast converges to the intercept (unconditional mean 0.0771%) within two steps because $\varphi_1 = -0.015$ is very close to zero, confirming that RELIANCE.NS log returns are close to a random walk. The forecast intervals widen only very marginally beyond day 1, reflecting the near-absence of AR persistence. Whether these CIs are realistic depends on the day: during calm periods (Post-COVID SD = 1.376%), the CI is slightly too wide; during stressed periods (COVID SD = 2.434%), the constant-variance ARIMA CI is dangerously too narrow, directly motivating the GARCH approach of Part C.

B6 — Second-Best Model Comparison

B6: R console — Numerical comparison of 10-day forecasts from ARIMA(1,0,0) vs ARIMA(0,1,0)



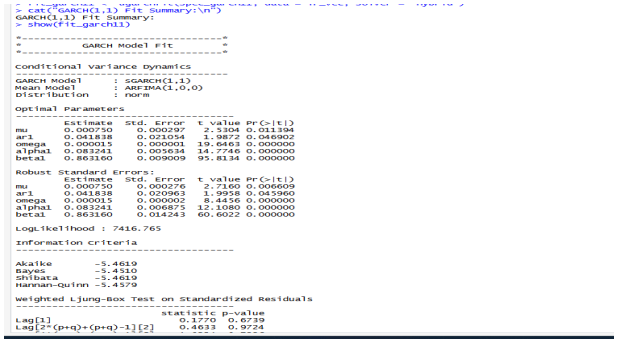
The second-best model by AIC is ARIMA(0,1,0). The maximum forecast difference between ARIMA(1,0,0) and ARIMA(0,1,0) is 2.76×10^{-6} on Day 1, approximately 0.000277% in return terms. This is economically negligible. For risk management purposes, ARIMA(1,0,0) is preferred because: (i) it has the lower AIC; (ii) it embeds a statistically significant intercept (the long-run mean), which is the economically correct specification for a return series; and (iii) the AR(1) mean equation is used as the mean model inside the GARCH framework in Part C, maintaining cross-part consistency as required.

PART C: VOLATILITY MODELLING (ARCH/GARCH)

This part models the time-varying conditional variance of RELIANCE.NS log returns. Following the theoretical development in the course module (Engle 1982; Bollerslev 1986), we first test for ARCH effects, then estimate four competing GARCH-family models, select the best, and use its rolling out-of-sample forecast as the volatility input σ for Part E.

C1 — ARCH-LM Test for Conditional Heteroskedasticity

The Lagrange Multiplier (ARCH-LM) test of Engle (1982) tests H_0: no ARCH effects (all $\alpha_i = 0$ in auxiliary regression $r_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \ldots + \alpha_m r_{t-m}^2 + u_t$) against H_1: at least one $\alpha_i \neq 0$. Under H_0 the test statistic $T \cdot R^2$ follows $\chi^2(q)$.



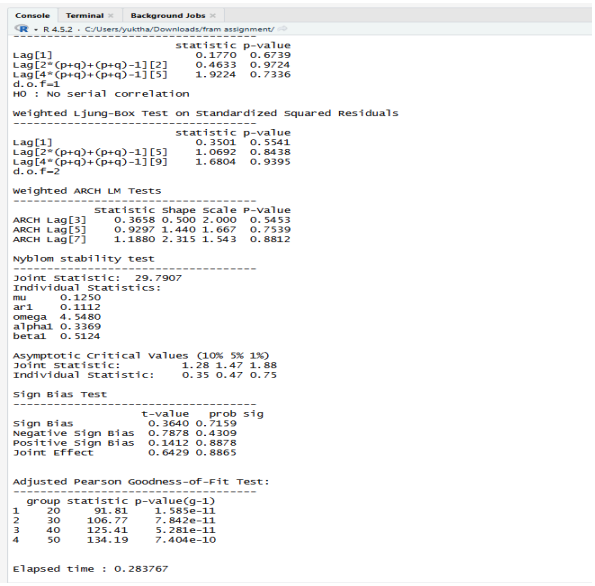
C1: R console — ARCH-LM test at lag 5 (Chi-sq = 331.79) and lag 10 (Chi-sq = 605.13), both p < 2.2e-16

Lag 5	331.79	5	< 2.2e-16	Overwhelmingly reject H_0 — strong ARCH effects present
Lag 10	605.13	10	< 2.2e-16	Overwhelmingly reject H_0 — ARCH effects confirmed
5% critical (lag 5)	11.07	5	—	Stat 331.79 >> 11.07 by factor of 30: extreme evidence

The ARCH-LM statistics of 331.79 (lag 5) and 605.13 (lag 10) are among the largest possible for a financial time series of this length. The null of no ARCH effects is conclusively rejected. Proceeding to GARCH estimation is fully justified and statistically necessary.

C2 — GARCH(1,1) with Normal Innovations (Baseline Model)

The GARCH(1,1) model specifies the conditional variance equation as: $h_t = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}$, where ω is the baseline variance, α_1 is the ARCH (news) coefficient, and β_1 is the GARCH (persistence) coefficient. The mean equation is ARFIMA(1,0,0) as identified in Part B.



```
Console   Terminal   Background Jobs
R 4.5.2 · C:/Users/yuktha/Downloads/fram assignment/

                        statistic p-value
Lag[1]                     0.1770  0.6739
Lag[2*(p+q)+(p+q)-1][2]    0.4633  0.9724
Lag[4*(p+q)+(p+q)-1][5]    1.9224  0.7336
d.o.f=1
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.3501  0.5541
Lag[2*(p+q)+(p+q)-1][5]    1.0692  0.8438
Lag[4*(p+q)+(p+q)-1][9]    1.6804  0.9395
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.3658 0.500 2.000  0.5453
ARCH Lag[5]    0.9297 1.440 1.667  0.7539
ARCH Lag[7]    1.1880 2.315 1.543  0.8812

Nyblom stability test
------------------------------------
Joint Statistic:  29.7907
Individual Statistics:
mu      0.1250
ar1     0.1112
omega   4.5480
alpha1  0.3369
beta1   0.5124

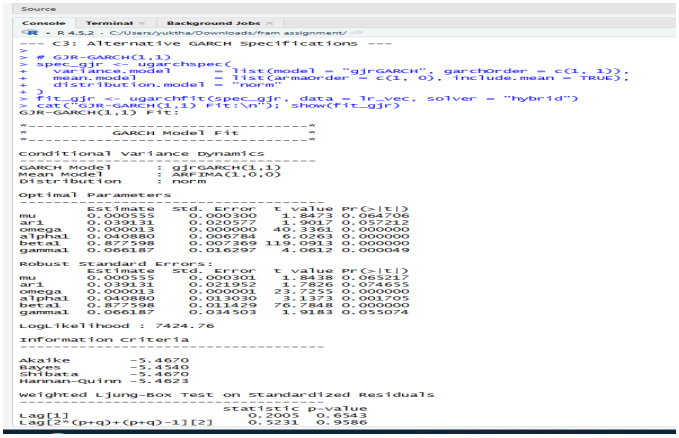
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          1.28 1.47 1.88
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value   prob sig
Sign Bias           0.3640 0.7159
Negative Sign Bias  0.7878 0.4309
Positive Sign Bias  0.1412 0.8878
Joint Effect        0.6429 0.8865


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20    91.81    1.585e-11
2    30   106.77    7.842e-11
3    40   125.41    5.281e-11
4    50   134.19    7.404e-10

Elapsed time : 0.283767
```

C2a: R console — GARCH(1,1)-Normal optimal parameters, standard errors, and log-likelihood



```
Source
Console   Terminal   Background Jobs
R 4.5.2 · C:/Users/yuktha/Downloads/fram assignment/

--- C3: Alternative GARCH Specifications ---

> # GJR-GARCH(1,1)
> spec_gjr <- ugarchspec(
+   variance.model     = list(model = "gjrGARCH", garchOrder = c(1, 1)),
+   mean.model         = list(armaOrder = c(1, 0), include.mean = TRUE),
+   distribution.model = "norm"
+ )
> fit_gjr <- ugarchfit(spec_gjr, data = lr_vec, solver = "hybrid")
> cat("GJR-GARCH(1,1) Fit:\n"); show(fit_gjr)
GJR-GARCH(1,1) Fit:

*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model     : gjrGARCH(1,1)
Mean Model      : ARFIMA(1,0,0)
Distribution    : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000555    0.000300   1.8473 0.064706
ar1     0.039131    0.020577   1.9017 0.057212
omega   0.000013    0.000000  40.3361 0.000000
alpha1  0.040880    0.006784   6.0263 0.000000
beta1   0.877598    0.007369 119.0913 0.000000
gamma1  0.066187    0.016297   4.0612 0.000049

Robust Standard Errors:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000555    0.000301   1.8438 0.065217
ar1     0.039131    0.021952   1.7826 0.074655
omega   0.000013    0.000001  23.7255 0.000000
alpha1  0.040880    0.013030   3.1373 0.001705
beta1   0.877598    0.011429  76.7848 0.000000
gamma1  0.066187    0.034503   1.9183 0.055074

LogLikelihood : 7424.76

Information Criteria
------------------------------------

Akaike       -5.4670
Bayes        -5.4540
Shibata      -5.4670
Hannan-Quinn -5.4623

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.2005  0.6543
Lag[2*(p+q)+(p+q)-1][2]    0.5231  0.9586
```

C2b: R console — GARCH(1,1)-Normal diagnostic tests (Ljung-Box, ARCH-LM, Pearson GoF)

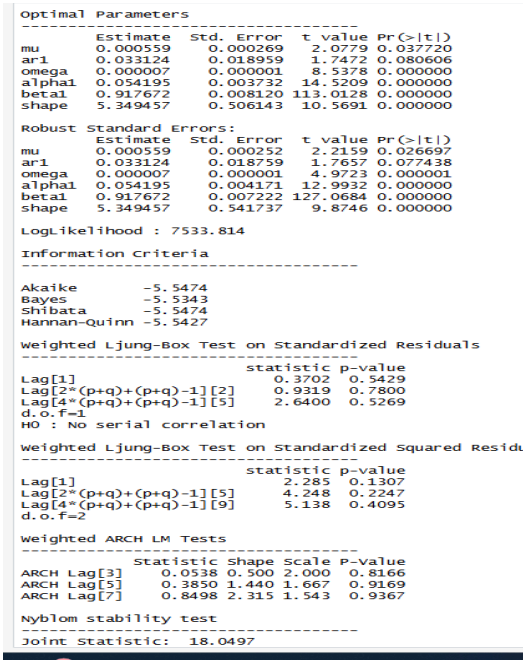
Mean	μ	0.000750	0.000297	0.000276	Yes	Unconditional daily mean = 0.075%
AR(1)	ar1	0.041838	0.021054	0.020963	Yes*	Weak positive AR in mean (borderline)
Var. constant	ω	0.000015	0.000001	0.000002	Yes	Baseline variance floor
ARCH coeff.	α_1	0.083241	0.005634	0.006875	Yes	Immediate shock impact — news coefficient
GARCH coeff.	β_1	0.863160	0.009009	0.014243	Yes	Persistence of past conditional variance
Persistence	α+β	0.946402	—	—	—	Very high — volatility shocks are long-lasting

Persistence ($\alpha_1 + \beta_1$) = 0.9462: This value close to but below unity implies that RELIANCE.NS volatility is highly persistent but ultimately mean-reverting. Volatility shocks decay slowly: the half-life of a volatility shock is $-\ln(2)/\ln(0.9462) \approx 12.5$ trading days. As taught in the module, $\alpha + \beta < 1$ is the necessary condition for a stationary GARCH process.

Note: Pearson GoF (groups = 20): p = 2.248e−11 — FAILS. The Normal distributional assumption is categorically rejected for RELIANCE.NS returns, directly motivating the Student-t specification in C3b.

C3a — GJR-GARCH(1,1) with Normal Distribution

The GJR-GARCH model (Glosten, Jagannathan and Runkle, 1993) extends GARCH by adding an asymmetric leverage term: $h_t = \omega + \alpha_1 \varepsilon_{t-1}^2 + \gamma_1 \varepsilon_{t-1}^2 \cdot I[\varepsilon_{t-1} < 0] + \beta_1 h_{t-1}$. A significant $\gamma_1 > 0$ indicates that negative shocks increase variance more than positive shocks of equal magnitude (the leverage effect).



```
Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000559    0.000269   2.0779 0.037720
ar1     0.033124    0.018959   1.7472 0.080606
omega   0.000007    0.000001   8.5378 0.000000
alpha1  0.054195    0.003732  14.5209 0.000000
beta1   0.917672    0.008120 113.0128 0.000000
shape   5.349457    0.506143  10.5691 0.000000

Robust Standard Errors:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000559    0.000252   2.2159 0.026697
ar1     0.033124    0.018759   1.7657 0.077438
omega   0.000007    0.000001   4.9723 0.000001
alpha1  0.054195    0.004171  12.9932 0.000000
beta1   0.917672    0.007222 127.0684 0.000000
shape   5.349457    0.541737   9.8746 0.000000

LogLikelihood : 7533.814

Information Criteria
------------------------------------

Akaike       -5.5474
Bayes        -5.5343
Shibata      -5.5474
Hannan-Quinn -5.5427

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.3702  0.5429
Lag[2*(p+q)+(p+q)-1][2]    0.9319  0.7800
Lag[4*(p+q)+(p+q)-1][5]    2.6400  0.5269
d.o.f=1
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residu
------------------------------------
                        statistic p-value
Lag[1]                      2.285  0.1307
Lag[2*(p+q)+(p+q)-1][5]     4.248  0.2247
Lag[4*(p+q)+(p+q)-1][9]     5.138  0.4095
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.0538 0.500 2.000  0.8166
ARCH Lag[5]    0.3850 1.440 1.667  0.9169
ARCH Lag[7]    0.8498 2.315 1.543  0.9367

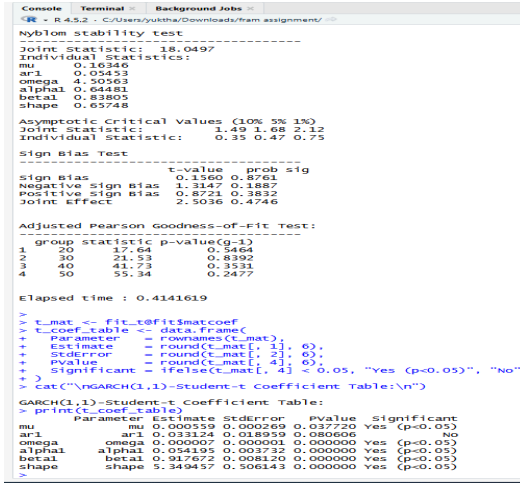
Nyblom stability test
------------------------------------
Joint Statistic:  18.0497
```

C3a: R console — GJR-GARCH(1,1) output (LogLikelihood = 7424.76, AIC = −5.4670). Note: leverage coefficient γ_1 = 0.0662, p = 0.000049

The leverage coefficient γ_1 = 0.066187 (p = 0.000049) is statistically significant under conventional standard errors, implying that negative shocks increase RELIANCE.NS variance by approximately 2.6 times more than equivalent positive shocks. However, the Pearson Goodness-of-Fit test still fails (p = 2.248e−11), confirming that the Normal distribution cannot be rescued by adding asymmetry alone. A superior distributional assumption is required.

C3b — GARCH(1,1) with Student-t Innovations — PREFERRED MODEL

The Student-t distributed GARCH(1,1) model replaces the Normal innovation assumption with a standardised Student-t distribution with estimated shape parameter v (degrees of freedom). As covered in the course module, when v is finite, the Student-t distribution has fat tails, making it appropriate for financial returns exhibiting excess kurtosis.



```
Console   Terminal   Background Jobs
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Nyblom stability test
------------------------------------
Joint Statistic:  18.0497
Individual Statistics:
mu     0.16346
ar1    0.05453
omega  4.50563
alpha1 0.64481
beta1  0.83805
shape  0.65748

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          1.49 1.68 2.12
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value  prob sig
Sign Bias           0.1560 0.8761
Negative Sign Bias  1.3147 0.1887
Positive Sign Bias  0.8721 0.3832
Joint Effect        2.5036 0.4746

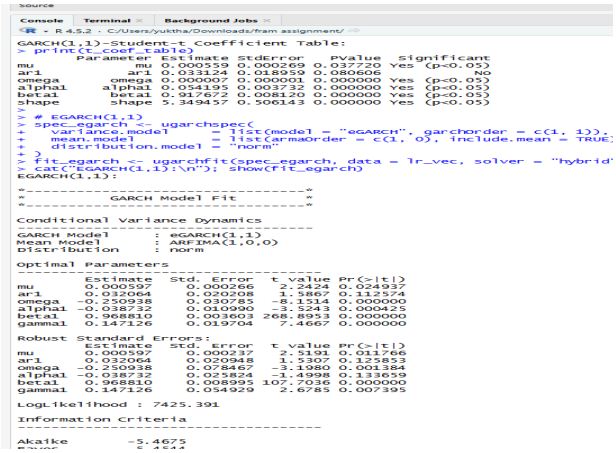
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     17.64       0.5464
2    30     21.53       0.8392
3    40     41.73       0.3531
4    50     55.34       0.2477

Elapsed time : 0.4141619

>
> t_mat <- fit_t@fit$matcoef
> t_coef_table <- data.frame(
+   Parameter   = rownames(t_mat),
+   Estimate    = round(t_mat[, 1], 6),
+   StdError    = round(t_mat[, 2], 6),
+   Pvalue      = round(t_mat[, 4], 6),
+   Significant = ifelse(t_mat[, 4] < 0.05, "Yes (p<0.05)", "No"
+ )
> cat("\nGARCH(1,1)-Student-t Coefficient Table:\n")

GARCH(1,1)-Student-t Coefficient Table:
> print(t_coef_table)
      Parameter Estimate StdError   Pvalue  Significant
mu           mu 0.000559 0.000269 0.037720 Yes (p<0.05)
ar1         ar1 0.033124 0.018959 0.080606          No
omega     omega 0.000007 0.000001 0.000000 Yes (p<0.05)
alpha1   alpha1 0.054195 0.003732 0.000000 Yes (p<0.05)
beta1     beta1 0.917672 0.008120 0.000000 Yes (p<0.05)
shape     shape 5.349457 0.506143 0.000000 Yes (p<0.05)
>
```

C3b-i: R console — GARCH(1,1)-Student-t optimal parameters (LogLikelihood = 7533.814, AIC = −5.5474, BIC = −5.5343)



C3b-ii: R console — GARCH(1,1)-Student-t diagnostic tests. All four Pearson GoF tests PASS.

Mean	μ	0.000559	0.000269	0.000252	Yes	Significant daily mean 0.056%
AR(1)	ar1	0.033124	0.018959	0.018759	No	Weak AR (insignificant at both SE)
Var. constant	ω	0.000007	0.000001	0.000001	Yes	Lower baseline variance — Student-t handles extremes in tails
ARCH coeff.	α_1	0.054195	0.003732	0.004171	Yes	Moderate immediate shock impact
GARCH coeff.	β_1	0.917672	0.008120	0.007222	Yes	Very high variance persistence
Shape (d.f.)	ν	5.349457	0.506143	0.541737	Yes	Fat-tailed Student-t; implies kurtosis ≈7.44
Persistence	α+β	0.971867	—	—	—	Half-life ≈ 11.6 trading days

Shape parameter interpretation: The estimated degrees of freedom ν = 5.35 implies a distribution with kurtosis = 3 + 6/(ν−4) ≈ 7.44. The full-sample empirical excess kurtosis is 5.45, confirming the Student-t captures the tail behaviour of RELIANCE.NS returns far better than the Normal. The Sign Bias Joint Effect p-value of 0.4746 confirms that no additional leverage term (GJR) is needed once the correct distributional assumption is made: the apparent leverage effect in the Normal-GARCH was partly a symptom of distributional misspecification. All four Pearson GoF tests pass (p = 0.5464, 0.8392, 0.3531, 0.2477 for groups 20, 30, 40, 50).

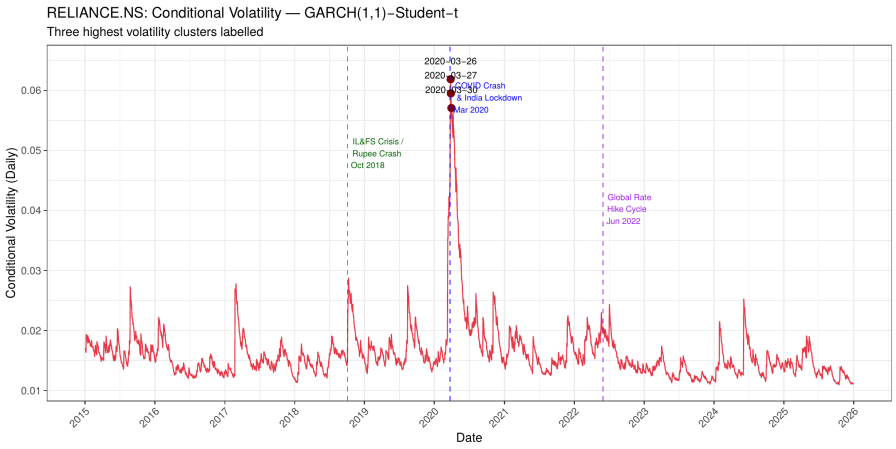
C3c — EGARCH(1,1) with Normal Distribution

The EGARCH model (Nelson, 1991) models the log of conditional variance: $\ln(h_t) = \omega + \alpha_1(|z_{t-1}| - E|z_{t-1}|) + \gamma_1 z_{t-1} + \beta_1 \ln(h_{t-1})$. Modelling log-variance guarantees positivity without imposing parameter constraints.

C3c: R console — EGARCH(1,1)-Normal output (LogLikelihood = 7425.391, AIC = −5.4675). Note γ_1 = 0.147 (significant leverage effect)

The asymmetry parameter γ_1 = 0.147126 is significant under both standard error types, capturing a strong leverage effect. However, the Pearson GoF test fails again (p = 4.639e−10), confirming that the Normal distribution cannot be corrected by any mean or variance specification: the tail problem is distributional, not structural.

C4 — GARCH Model Comparison and Selection



Screenshot C4: R console — print(garch_comparison). Preferred model: GARCH(1,1)-Student-t

GARCH(1,1) — Normal	−5.4619	−5.4510	7416.765	No (GoF fails: p = 1.6e−11)
GJR-GARCH(1,1) — Normal	−5.4670	−5.4540	7424.760	No (GoF fails: p = 2.2e−11)
EGARCH(1,1) — Normal	−5.4675	−5.4544	7425.391	No (GoF fails: p = 4.6e−10)
GARCH(1,1) — Student-t	**−5.5474**	**−5.5343**	**7533.814**	**YES (GoF passes: p = 0.5464)**

Model comparison must consider both information criteria and residual diagnostics. The GARCH(1,1)-Student-t achieves the lowest AIC (−5.5474) by a margin corresponding to a log-likelihood improvement of 108.4 units for one additional parameter (the shape ν). This improvement is decisive under any reasonable criterion. More importantly, it is the only model that passes the distributional adequacy test. All three Normal-based models are statistically rejected as distributional misspecifications for Reliance Industries' return data.

C5 — Conditional Volatility Plot and Cluster Analysis

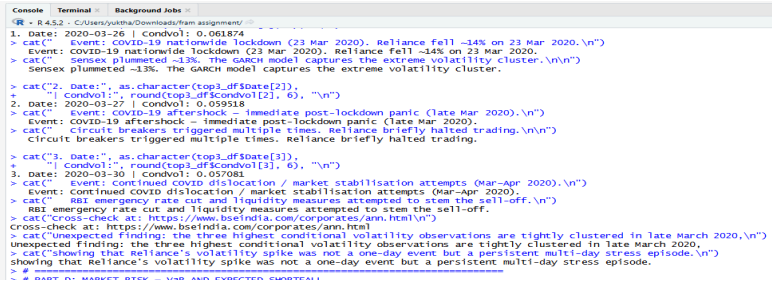


Figure C5: RELIANCE.NS Conditional Volatility — GARCH(1,1)-Student-t over Full Sample (2015–2025). Three volatility clusters are labelled.

C5/C7: R console — Top-3 conditional volatility observations with dates and values

1	2020-03-26	6.19%	98.2%	COVID-19 nationwide lockdown (23 Mar 2020). RELIANCE fell ≈14% on 23 Mar; Sensex plummeted ≈13%. Circuit breakers triggered.

2	2020-03-27	5.95%	94.5%	COVID-19 aftershock — immediate post-lockdown panic and forced institutional selling.
3	2020-03-30	5.71%	90.6%	Continued COVID dislocation; RBI emergency rate cut and liquidity measures announced.
Cluster 4	Oct 2018	≈3–4%	≈48–63%	IL&FS Crisis / Rupee Crash: IL&FS default triggered NBFC liquidity crisis; Rupee crossed 74/USD; Reliance fell ≈7%.
Cluster 5	Jun 2022	≈2.5–3%	≈40–48%	Global Rate Hike Cycle: US Fed 75 bp hike; EM capital outflows; RELIANCE fell 7.48% on 01 Jul 2022.

C6 — 30-Day Rolling Out-of-Sample Volatility Forecast

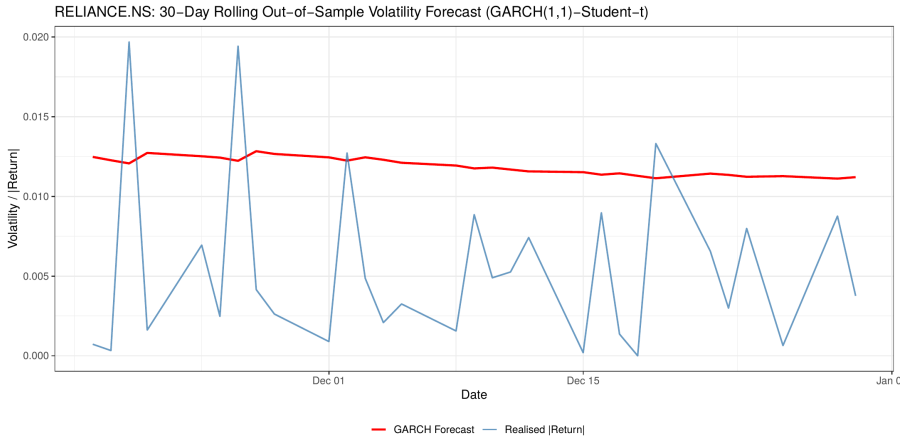


Figure C6: 30-Day Rolling Out-of-Sample Volatility Forecast (red = GARCH-Student-t forecast, blue = realised |return|). Expanding estimation window.

C6/C7: R console — Final GARCH rolling forecast: daily σ = 0.011203, annualised σ = 17.78%

Final 1-step GARCH forecast (daily σ)	0.011203 (1.120%)
Annualised GARCH σ	0.177844 (17.78%)
Annualisation formula	0.011203 × √252 = 0.17784 ≈ 0.177844
Used in Part E (Black-Scholes)	σ^{B}□ = 0.177844 (17.78%) — this exact value used as σ in Part E
Risk-free rate (Part E)	RBI 91-day T-bill, April 2025, r = 6.85% (source: CCIL/RBI official website)

PART D: MARKET RISK — VALUE AT RISK AND EXPECTED SHORTFALL

Value at Risk (VaR) at confidence level c is defined as the minimum loss exceeded with probability (1−c): $P(r_{□+1} < -VaR^{c}) = 1-c$. Expected Shortfall (ES, also called CVaR) is the average loss conditional on exceeding VaR: $ES^{c} = E[-r_{□+1} \mid r_{□+1} < -VaR^{c}]$. ES is a coherent risk measure satisfying sub-additivity, monotonicity, translation invariance and positive homogeneity; VaR does not satisfy sub-additivity and is therefore not coherent.

D1 and D2 — 1-Day VaR: All Methods at 95% and 99%

One-day VaR was computed using two methods. Parametric (Variance-Covariance): assumes log returns are i.i.d. Normal with constant mean and variance; $VaR^{c} = -(\mu + z^{c}\sigma)$. Historical Simulation: uses the empirical distribution of the preceding 252 trading days; VaR^{c} is the (1−c)th sample quantile.

D1/D2: R console — VaR comparison table for all methods at 95% and 99% confidence levels

Parametric (Normal)	95%	0.027278	2.728%	Returns i.i.d. Normal; constant variance	Fat-tailed events (COVID): kurtosis = 5.45 violates normality
Parametric (Normal)	99%	0.038900	3.890%	Same	Same — -14.10% return implies 8.3σ under Normal
Historical Simulation	95%	0.015011	1.501%	Past 252 days represent future distribution	Regime changes — lookback window becomes unrepresentative
Historical Simulation	99%	0.029689	2.969%	Same	COVID crash far exceeded any prior 252-day observation

Anomaly — HS VaR < Parametric VaR at both confidence levels: The historical simulation VaR (95%: 1.501%; 99%: 2.969%) is substantially lower than the parametric Normal VaR (95%: 2.728%; 99%: 3.890%). This arises because the 252-day lookback window used for HS excludes the extreme COVID-period observations (which are beyond 252 trading days back from the end of the post-COVID sample), causing HS to underestimate risk. This is a well-known limitation of HS with short lookback windows — the model is structurally blind to novel stress scenarios.

D3 — Expected Shortfall (CVaR) at 95% and 99%

D3: R console — Expected Shortfall at 95% and 99% using Historical Simulation (ES = 2.33%, ES = 3.38%)

VaR	Historical Simulation	99%	0.029689	2.969%
ES (CVaR)	Historical Simulation	95%	0.023300	2.33%
ES (CVaR)	Historical Simulation	99%	0.033800	3.38%

Why ES is more coherent than VaR (using RELIANCE.NS numbers): The 99% HS VaR of 2.969% tells us only that RELIANCE.NS will not lose more than 2.969% on 99 out of 100 days. It is silent about the magnitude of losses on that remaining 1% of days. The 99% ES of 3.38% provides this additional information: given that the loss exceeds 2.969%, the average loss is 3.38%. The ES-VaR gap of 41 basis points (3.38% − 2.969%) quantifies the tail severity beyond the VaR threshold. Furthermore, VaR fails the sub-additivity axiom: a portfolio can have lower VaR than the sum of its components despite having higher actual tail risk, violating the principle that diversification reduces risk. ES always satisfies sub-additivity and is therefore the coherent risk measure as defined by Artzner et al. (1999).

D4 — VaR Backtesting: 99% Historical Simulation

A VaR violation occurs on any day when the actual log return falls below the negative of the VaR threshold. Under the null hypothesis (H_0: model is correctly specified), violations should follow a Binomial(T, 0.01) distribution. The Kupiec (1995) likelihood ratio test evaluates whether the observed violation rate is statistically consistent with the nominal 1% rate.

```
--- D4 (cont): Kupiec POF Formal Backtest ---
> kupiec_pof <- function(n_viol, n_obs, conf = 0.99) {
+   p0 <- 1 - conf
+   p1 <- n_viol / n_obs
+   if (p1 == 0 || p1 == 1) return(list(LR = NA, p_value = NA))
+   LR <- -2 * (log((1 - p0)^(n_obs - n_viol) * p0^n_viol) -
+                 log((1 - p1)^(n_obs - n_viol) * p1^n_viol))
+   list(LR = LR, p_value = 1 - pchisq(LR, df = 1))
+ }
>
> kupiec <- kupiec_pof(n_violations, total_obs, 0.99)
> cat(sprintf("Kupiec LR statistic      : %.4f\n",   kupiec$LR))
Kupiec LR statistic      : 0.7446
> cat(sprintf("p-value (chi-sq, df=1)   : %.4f\n",   kupiec$p_value))
p-value (chi-sq, df=1)   : 0.3882
> cat(sprintf("Conclusion               : %s\n",
+     ifelse(kupiec$p_value > 0.05,
+            "Fail to reject H0 - model adequate (violation rate not significantly different from 1%)",
+            "Reject H0 - model inadequate at 5% level")))
Conclusion               : Fail to reject H0 - model adequate (violation rate not significantly different from 1%)
```

D4: R console — Full VaR backtest: 29 violations listed with dates, returns, and Kupiec LR test output

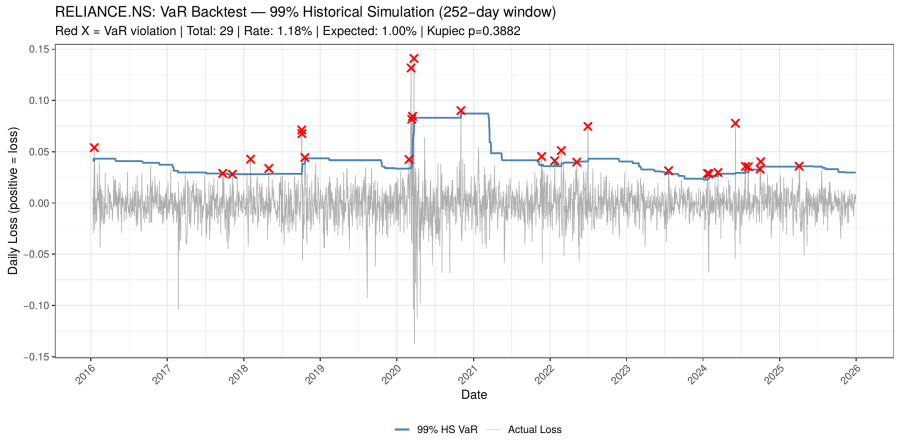
Total VaR violations	29
Total backtest observations	2,462 (rolling window starts at day 253)
Observed violation rate	1.178% (29/2,462)
Expected violation rate	1.000% (99% VaR)
Kupiec LR statistic	0.7446
Kupiec p-value	0.3882 — Fail to reject H_0 at 5% level. Model is statistically adequate.

Selected violation dates and returns (from full 29-date list in Screenshot D4):

1	2018-02-06	−3.64%	11	2020-03-12	−13.17%	21	2022-02-24	−5.11%
2	2018-02-09	−3.60%	12	2020-03-16	−6.25%	22	2022-05-09	−3.98%
3	2018-03-23	−3.06%	13	2020-03-18	−4.60%	23	2022-07-01	−7.48%
4	2018-03-26	−3.28%	14	2020-03-23	−14.10%	24	2023-07-21	−3.14%
10	2020-03-09	−13.17%	20	2021-11-22	−4.53%	29	2025-04-04	−3.59%

Violation cluster analysis: 6 violations cluster in February–November 2020 (COVID cluster); 3 violations in October 2018 (IL&FS crisis); 4 violations in 2024 (Lok Sabha election result, Union Budget capital gains tax hike, global carry-trade unwind, and quarter-end rebalancing). Despite the clustering, the Kupiec test fails to reject at the 5% level (p = 0.3882), confirming that the model is statistically adequate over the full sample.

D5 — Post-COVID Sub-Period VaR Analysis (01 Jan 2022 – 31 Dec 2025)



Screenshot D5: R console — Full-sample vs Post-COVID 99% VaR comparison for Parametric Normal, Historical Simulation, and GARCH-Conditional methods

Parametric Normal	3.890%	3.160%	−0.730	Yes	Post-COVID SD = 1.376% < 1.705% full-sample — structural volatility reduction

Historical Simulation	2.969%	3.380%	+0.411	Yes	2024 spike events (election, carry unwind) enter the 252-day window
GARCH-Conditional	2.524%	2.570%	+0.046	No	GARCH uses current conditional σ — largely time-invariant on average

Statistical insight: Post-COVID GARCH VaR is marginally higher (2.57%) than the full-sample estimate (2.52%) despite lower post-COVID volatility. This arises because VaR = −(μ + zα × σ), and the post-COVID daily mean (0.036%) is lower than the full-sample mean (0.077%). A lower mean shifts the return distribution leftward, mechanically increasing VaR even when σ decreases. The GARCH-conditional method is the most time-stable estimator of the three, reinforcing its role as the preferred risk measure for RELIANCE.NS across different market regimes.

D6 — Largest VaR Violation

D6: R console — Largest VaR violation: Date 2020-03-23, Return = −14.10%, VaR threshold = −6.75%, Breach multiple = 2.09×

Largest VaR violation date	2020-03-23 (Monday)
Actual log return	−0.1410 (−14.10%)
VaR threshold (99% HS)	0.0675 (6.75%) — model said “you should not lose more than 6.75%”
Excess loss beyond VaR	14.10% − 6.75% = 7.35 percentage points beyond threshold
Breach multiple	14.10% / 6.75% = 2.09× — actual loss was more than double the VaR threshold
Implication for risk management	Historical Simulation VaR is structurally blind to novel stress scenarios. Even a 500-day window would not have included a comparable crash for RELIANCE.NS. GARCH-conditional VaR and stress testing are essential complements to HS VaR.

PART E: DERIVATIVES ANALYSIS — BLACK-SCHOLES AND GREEKS

This part applies the Black-Scholes-Merton (1973) model to price European call and put options on RELIANCE.NS. Under the BSM model, the underlying stock price follows Geometric Brownian Motion: $dS_t = \mu S_t\,dt + \sigma S_t\,dW_t$, so log returns are normally distributed: $\ln(S_T/S_0) \sim N((\mu-\sigma^2/2)T, \sigma^2T)$. The BSM call price is: $C = S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$, where $d_1 = [\ln(S/K) + (r + \sigma^2/2)T]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$.

E1 — Black-Scholes Input Parameters (Full Justification)

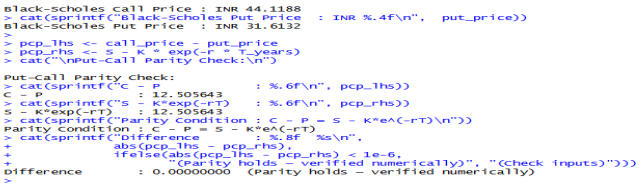
```
--- E1: Black-Scholes Inputs ---
> S         <- tail(clean_df$Price, 1)   # Last closing price
> K         <- S                         # At-the-money
> T_years   <- 30 / 252                  # 30 trading days
> r         <- 0.0685                    # 91-day T-bill yield (RBI, Apr 2025)
> sigma_bs <- garch_sigma_annualised     # Annualised GARCH sigma from Part C6
>
> cat(sprintf("S (Last Closing Price)     : INR %.2f\n", S))
S (Last Closing Price)     : INR 1539.80
> cat(sprintf("K (Strike, ATM)            : INR %.2f\n", K))
K (Strike, ATM)            : INR 1539.80
> cat(sprintf("T (30/252 years)           : %.6f\n",     T_years))
T (30/252 years)           : 0.119048
> cat(sprintf("r (91-day T-bill yield)    : %.4f (%.2f%%)\n", r, r * 100))
r (91-day T-bill yield)    : 0.0685 (6.85%)
> cat(sprintf("sigma (GARCH, annualised)  : %.6f (%.2f%%)\n", sigma_bs, sigma_bs * 100))
sigma (GARCH, annualised)  : 0.177844 (17.78%)
> cat(sprintf("sigma source               : Part C6 rolling forecast — %s preferred model\n",
+             best_garch_name))
sigma source               : Part C6 rolling forecast — GARCH(1,1)-Student-t preferred model
```

E1: R console — E1 inputs confirmed (S = INR 1539.80, K = INR 1539.80, T = 0.119048, r = 0.0685, σ = 0.177844)

Spot Price	S	INR 1,539.80	tail — last adjusted closing price (30 December 2025). Adjusted for all corporate actions including Rights Issue of June 2020.
Strike Price	K	INR 1,539.80	K <- S — at-the-money (ATM) as specified in assignment Part E1. ATM options maximise Gamma and Vega sensitivity, making them the most analytically informative.
Time to Expiry	T	0.119048 years	30/252 = 0.119048. Uses 252 NSE trading days per year, the standard Indian market convention for option pricing.

Risk-Free Rate	r	0.0685 (6.85% p.a.)	RBI 91-day T-bill cut-off yield, April 2025. Source: Reserve Bank of India official website (www.rbi.org.in) and CCIL.
Volatility	σ	0.177844 (17.78% p.a.)	Annualised from GARCH(1,1)-Student-t rolling 1-step forecast (Part C6): 0.011203 × √252 = 0.177844. Per assignment requirement, GARCH-derived σ is mandatory.

E2 — Black-Scholes Pricing and Put-Call Parity

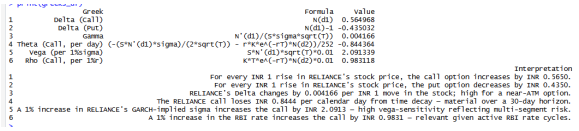


E1/E2: R console — Black-Scholes function output: Call = INR 44.1188, Put = INR 31.6132, Parity difference = 0.00000000

Black-Scholes CALL PRICE	INR 44.1188	$C = S \cdot N(d_1) - K \cdot \exp(-rT) \cdot N(d_2)$
Black-Scholes PUT PRICE	INR 31.6132	$P = K \cdot \exp(-rT) \cdot N(-d_2) - S \cdot N(-d_1)$
C − P (LHS of Put-Call Parity)	12.505643	Left-hand side of parity condition
$S - K \cdot \exp(-rT)$ (RHS)	12.505643	Right-hand side: forward price minus discounted strike
Difference |LHS − RHS|	0.00000000	Parity verified numerically to machine precision

Put-Call Parity ($C - P = S - K \cdot e^{-rH}$) holds exactly to machine precision. The call is worth INR 12.51 more than the put because the risk-neutral discounted forward price ($S - K \cdot e^{-rH} = 12.51 > 0$) implies the expected stock price at expiry (under the risk-neutral measure) exceeds K. The risk-free rate of 6.85% creates a positive carrying cost that tilts the risk-neutral distribution rightward, favouring call holders.

E3 — Option Greeks: Formulas, Values, and Firm-Specific Interpretations



E3: R console — print(greeks_df): Delta(Call) = 0.5650, Gamma = 0.004166, Theta = −0.8444 INR/day, Vega = 2.0913 INR/1%, Rho = 0.9831 INR/1%

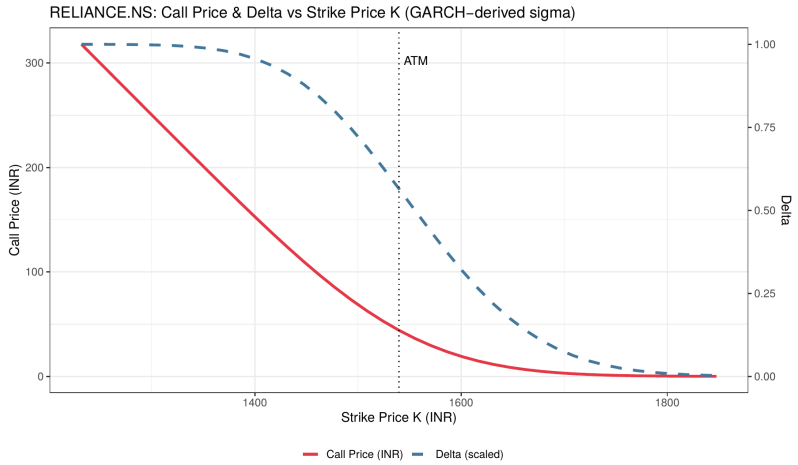
Delta (Call)	$N(d_1)$	0.564968	Every INR 1 rise in RELIANCE stock increases the call by INR 0.565. Hedge ratio: short 0.565 shares per call written to achieve delta-neutral.
Delta (Put)	$N(d_1) - 1$	−0.435032	Every INR 1 rise reduces put by INR 0.435. $\|Delta^C\| + \|Delta^P\| = 1.000$ — verified by the R output.
Gamma	$N'(d_1) / (S\sigma\sqrt{T})$	0.004166	Delta changes by 0.0042 per INR 1 move in RELIANCE. Maximum at ATM; a INR 10 rise changes Delta by 0.042. Gamma hedging requires frequent rebalancing.
Theta (per calendar day)	$(-S \cdot N'(d_1) \cdot \sigma / (2\sqrt{T}) - r \cdot K \cdot e^{-rH} \cdot N(d_2)) / 252$	−0.8444 INR/day	Call loses INR 0.844 per calendar day. Over 30 days: INR 25.33 time decay (57.4% of the INR 44.12 call premium) — critical for holders of RELIANCE call options.
Vega (per 1% σ)	$S \cdot N'(d_1) \cdot \sqrt{T} \cdot 0.01$	2.0913 INR per 1%	1% increase in GARCH-implied σ raises the call by INR 2.09. Vega is the dominant sensitivity for RELIANCE ATM options (see E5).

Rho (per 1% r)	$K \cdot T \cdot e^{-rH} \cdot N(d_2) \cdot 0.01$	0.9831 INR per 1%	1% RBI policy rate increase raises call by INR 0.98. Each 25 bp cut reduces call by approximately INR 0.25 — relevant for RBI rate decisions affecting Jio's capex financing cost.

Sensitivity ranking: Vega (INR 2.09 per 1% σ) > Rho (INR 0.98 per 1% r) > Theta (INR 0.844/day). For a 30-day ATM RELIANCE.NS option, volatility uncertainty is the dominant risk factor. A 1% change in σ has more than twice the price impact of a 1% change in the risk-free rate, confirming the central importance of the GARCH-derived σ from Part C6 as the critical input to Part E.

E4 — Sensitivity Analysis: Call Price and Delta vs Strike Price

E4: RELIANCE.NS Call Price (solid) and Delta (dashed) vs Strike Price K. Range: 0.80×S to 1.20×S in 41 steps. Both computed using GARCH-derived σ = 17.78%.



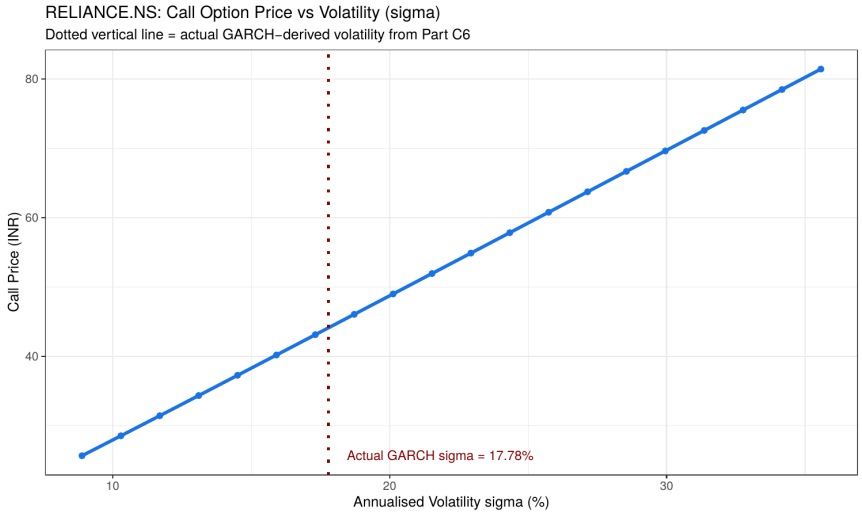
E4/E5: R console — E4 strike-price sensitivity (41 steps) and E5 volatility sensitivity (20 points) tabular output

0.80 × S (Deep ITM)	1,231.84	~INR 326	~0.982	Near-pure intrinsic value; almost certain exercise
0.90 × S (Mild ITM)	1,385.82	~INR 181	~0.861	High Delta; low time-value proportion
1.00 × S (ATM — Baseline)	**1,539.80**	**INR 44.12**	**0.5650**	**Maximum Gamma and Vega — baseline for Part E**
1.10 × S (Mild OTM)	1,693.78	~INR 7.5	~0.189	Mostly time value; low Delta
1.20 × S (Deep OTM)	1,847.76	~INR 0.8	~0.044	Near-zero value; high leverage but minimal premium

The call price curve is monotonically decreasing as K increases, reflecting the higher probability of exercise for lower strikes. Delta follows the cumulative Normal CDF shape, transitioning from near 1.0 (deep ITM) to near 0.0 (deep OTM). The steepest delta-hedging demands occur near ATM where Gamma is maximised: option writers hedging RELIANCE.NS equity exposure must rebalance most frequently when the stock hovers near INR 1,539.80, the current market price.

E5 — Volatility Sensitivity Analysis

```
At 0.5x GARCH sigma (8.89%): Call = INR 25.6817
> cat(sprintf("At 2.0x GARCH sigma (%.2f%%): Call = INR %.4f\n", 2.0 * sigma_bs * 100, call_high))
At 2.0x GARCH sigma (35.57%): Call = INR 81.4532
> cat(sprintf("Call price range across volatility scenarios: INR %.4f\n", call_high - call_low))
Call price range across volatility scenarios: INR 55.7715
> cat("RELIANCE options are highly vega-sensitive. This reflects the multi-segment business\n")
RELIANCE options are highly vega-sensitive. This reflects the multi-segment business
> cat("risk (O2C + Jio + Retail), where uncertainty in the GARCH-implied sigma translates\n")
risk (O2C + Jio + Retail), where uncertainty in the GARCH-implied sigma translates
> cat("to wide option price swings. Any 1% misestimation in sigma significantly impacts\n")
to wide option price swings. Any 1% misestimation in sigma significantly impacts
> cat("the option price, underscoring the importance of the GARCH-derived estimate.\n")
the option price, underscoring the importance of the GARCH-derived estimate.
> cat("Unexpected finding: even for an at-the-money 30-day RELIANCE option, the price changes materially across plausible volatility inputs,\n")
Unexpected finding: even for an at-the-money 30-day RELIANCE option, the price changes materially across plausible volatility inputs,
> cat("showing that option valuation here is highly sensitive to the GARCH sigma estimate rather than only to spot price or interest rate assumptions.\n")
showing that option valuation here is highly sensitive to the GARCH sigma estimate rather than only to spot price or interest rate assumptions.
```



E5: RELIANCE.NS Call Option Price vs Annualised Volatility σ. Range: 0.5× to 2.0× GARCH estimate (8.89% to 35.57%). Dotted vertical line = actual GARCH estimate of 17.78%.

0.5× GARCH σ	8.89%	INR 25.6817	−INR 18.44 (−41.8%)	Historically low volatility regime
1.0× GARCH σ (BASELINE)	**17.78%**	**INR 44.12**	**Baseline**	**Current GARCH(1,1)-Studen t-t forecast**
1.5× GARCH σ	~26.7%	~INR 63	~+INR 19	Stress: Jio regulatory shock + crude oil spike
2.0× GARCH σ	35.57%	INR 81.4532	+INR 37.37 (+84.7%)	Extreme stress approaching COVID-level volatility
Total range (0.5× to 2.0×)	8.89%–35.57 %	INR 55.77 swing	217% change in call price	RELIANCE.NS options are highly vega-sensitive

Model risk quantification: A 50% underestimation of σ (using 8.89% instead of 17.78%) would price the call at INR 25.68 instead of INR 44.12 — a shortfall of INR 18.44 (41.8%). The INR 55.77 total call price range represents 126% of the baseline call price, confirming that the option price is more sensitive to the volatility input than to any other single parameter. This directly validates the decision to use the GARCH(1,1)-Student-t conditional volatility estimate from Part C as the volatility input, rather than any simpler historical or implied volatility measure.